

Ros Driver Lidar

We provide the lidar ROS driver package, which has been successfully compiled and can drive the lidar on the Orin-Nano motherboard. Orin-Nano motherboard system environment: **Ubuntu22.04+ROS2-Humble**.

Installing Dependencies

Enter the following command in the terminal to install the necessary dependency files,

```
sudo apt-get update  
sudo apt-get install -y libyaml-cpp-dev libpcap-dev
```

Transplanting the Package and Compiling

Copy the contents of [ROS_SDK] from the Attachments folder to your workspace directory, then go back to the workspace and enter the following command to compile,

```
colcon build
```

Wait for the compilation to pass.

Lidar Connection

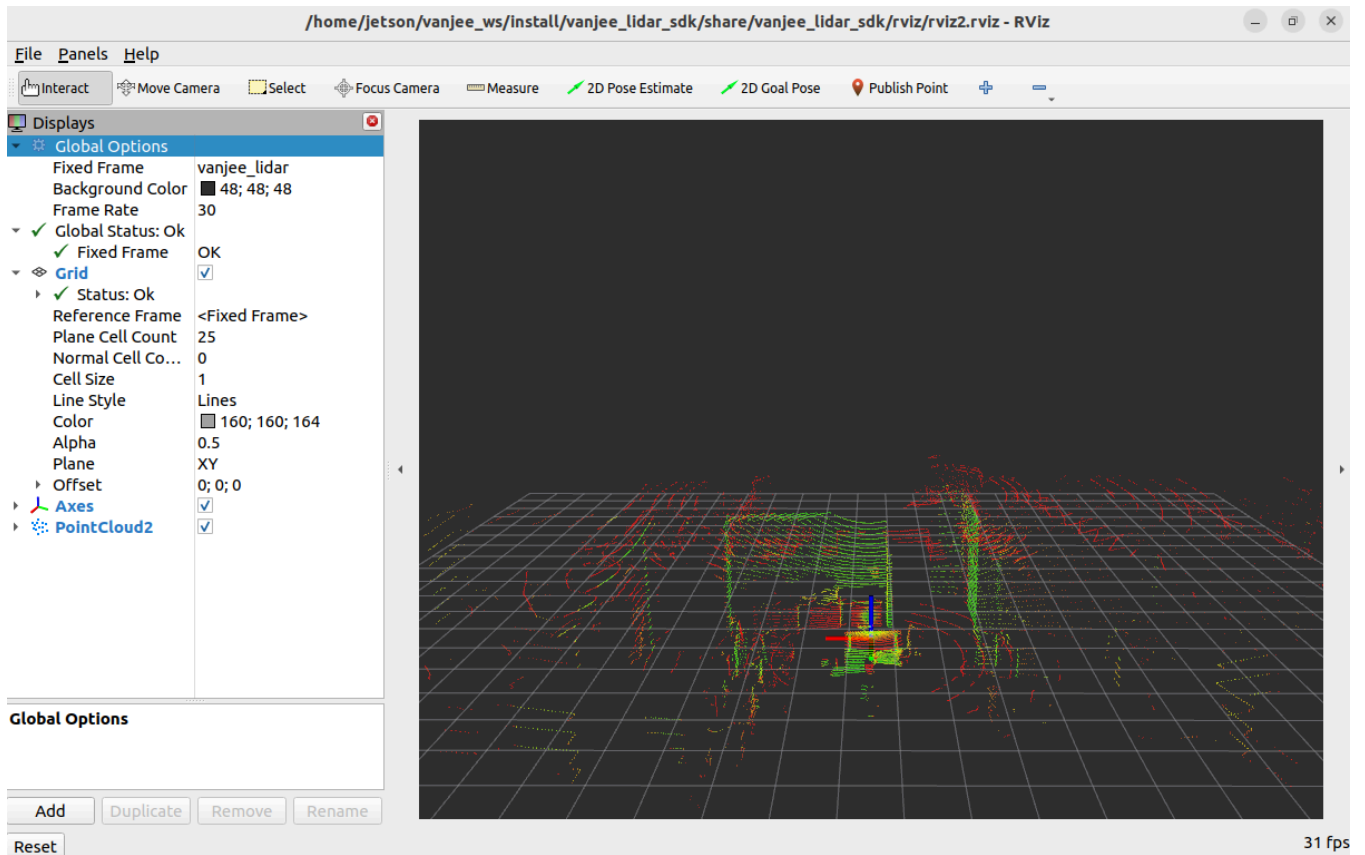
Connect the lidar to the lidar adapter cable, making sure to align the raised and recessed parts of the interface to avoid inserting it backwards and damaging the interface. Then power the lidar adapter cable's power port, and connect the Ethernet port to the motherboard's Ethernet port.

Driving the Lidar

Open a terminal and enter the lidar startup command,

```
ros2 launch vanjee_lidar_sdk start.py
```

After running, an rviz will open, which will display the 32-line lidar point cloud,



Viewing Topics

To view the lidar point cloud topic, enter in the terminal,

```
ros2 topic echo /rslidar_points
```

To view the imu topic, enter in the terminal,

```
ros2 topic echo /vanjee_lidar_imu_packets
```